

Logical Programming Practice Set (90 Problems)

1. Pattern Printing

1. Hollow Square Pattern
2. Floyd's Triangle
3. Diamond Star Pattern
4. Number Pyramid
5. Butterfly Pattern

2. Number Logic

1. Reverse a number without converting to string
2. Armstrong Number
3. Strong Number
4. Perfect Numbers in range
5. GCD and LCM

3. Prime Number Logic

1. Check Prime
2. First N Primes
3. Twin Primes
4. Prime Factors
5. Count Primes in Range

4. Digit Based Problems

1. Sum of digits
2. Product of digits
3. Largest digit
4. Smallest digit
5. Digit frequency

5. Loop Logic

1. Factorial
2. Tables 2-20
3. Fibonacci (N terms)
4. Fibonacci ($< N$)
5. Nth Fibonacci

6. Nested Loop Logic

1. Multiplication Matrix

2. Chessboard Pattern
3. Pascal Triangle
4. Spiral Pattern
5. Zigzag Pattern

7. String Logic

1. Reverse String
2. Palindrome
3. Count vowels, consonants, digits and spaces
4. Remove duplicates
5. String Compression

8. List Logic

1. Second Largest
2. Remove Duplicates
3. Rotate by K
4. Missing Number
5. Move Zeros to End

9. Searching Logic

1. Linear Search
2. Binary Search
3. Count Occurrences
4. First Occurrence
5. Last Occurrence

10. Sorting Logic

1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Sort without sort()
5. Even before Odd

11. Dictionary Logic

1. Word Frequency
2. Character Frequency
3. Max Occurring Character
4. Merge Dictionaries
5. Invert Dictionary

12. Function Logic

1. Calculator using Functions
2. Perfect Number Function
3. Recursive Factorial
4. Recursive Fibonacci
5. Recursive Sum of Digits

13. Recursion

1. Power(x,n)
2. Decimal to Binary
3. Reverse String
4. Tower of Hanoi
5. Generate Subsets

14. Matrix Logic

1. Matrix Addition
2. Matrix Multiplication
3. Transpose
4. Symmetric Matrix
5. Rotate 90 Degrees

15. Miscellaneous

1. Decimal to Binary
2. Binary to Decimal
3. Decimal to Octal
4. Decimal to Hexadecimal
5. Trailing Zeros in Factorial

16. Moderate Challenges

1. Happy Numbers
2. Pythagorean Triplets
3. Automorphic Numbers
4. Strong Numbers in Range
5. Longest Consecutive Increasing Sequence

Final Boss Level

1. Sudoku Validator
2. Tic-Tac-Toe Winner Checker
3. Matrix Spiral Traversal

4. Snake Pattern Matrix
5. Rotate Matrix Layer by Layer
6. Caesar Cipher
7. Run-Length Encoding
8. Roman to Integer and Integer to Roman Converter
9. Calendar Generator
10. Mini ATM Simulator

7-Day Practice Roadmap

- Day 1: Pattern Printing + Number Logic
- Day 2: Prime + Digit Problems
- Day 3: Loop + Nested Loop
- Day 4: String + List
- Day 5: Searching + Sorting + Dictionary
- Day 6: Functions + Recursion + Matrix
- Day 7: Moderate Challenges + Final Boss